Q U A N T U M L A T T I C E M O D E L

Reduced-Action Foundations and Planck-Unit Derivations

Quinton R. D. Tharp

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Abstract

This paper develops the Planck-unit foundations of the Quantum Lattice Model (QLM), a phase–action reformulation of natural units in which the Planck scale is re-expressed at the *per-radian* level rather than through dimensional square-root combinations of $\{G, \hbar, c\}$.

The QLM begins from the covariant primitive phase–flow law

$$E = \hbar \frac{d\theta}{d\tau},$$

where θ is dimensionless phase and τ is proper time along a worldline. Energy is therefore defined as reduced action flow per unit proper time.

The fundamental Planck energy arises as the saturated single-tick realization of this primitive relation:

$$\Delta\theta = 1, \qquad \Delta\tau = t_P \quad \Rightarrow \quad E_P = \frac{\hbar}{t_P}.$$

The expression $E_P = \hbar/t_P$ is thus not an independent definition, but the single-tick realization of the per-radian phase–flow rule.

When expressed in terms of the primitive triplet $(\hbar, \ell_P, t_P)$, all Planck quantities reduce to algebraically minimal forms. Mechanical, electromagnetic, geometric, and thermodynamic units admit a unified representation within the same phase–action structure, with examples including

$$m_P = \frac{\hbar t_P}{\ell_P^2}, \qquad p_P = \frac{\hbar}{\ell_P}.$$

The Bohr identity

$$\hbar = m_e v_B a_0$$

appears naturally within the same framework, showing that hydrogenic structure provides an exact algebraic bridge between atomic angular momentum and the Planck-scale action quantum, without introducing additional dimensional parameters.

Electromagnetic quantities—including the Planck charge, current, voltage, power, and impedance—emerge from the same substitutions. In the per-radian normalization, the Planck impedance is

$$Z_P = \frac{Z_0}{4\pi},$$

with the corresponding loop-aggregated form

$$Z_P^{(\mathrm{loop})} = 2\pi Z_P = \frac{Z_0}{2}, \qquad Z_P = \frac{Z_P^{(\mathrm{loop})}}{2\pi}.$$

Through symbolic derivations, dimensional analyses, and CODATA-validated numerical evaluations, this work establishes the complete per-radian Planck-unit system on which the broader Quantum Lattice Model is constructed. It provides the formal Planck-scale reference structure for subsequent developments in gravitational impedance, invariant transport and Lorentz kinematics, and phase-coherent dynamical structure.

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1 Introduction

Planck units are conventionally introduced as dimensional constructs formed from the constants $\{\hbar, G, c\}$, yielding expressions such as $\sqrt{\hbar c^5/G}$ and $\sqrt{\hbar c/G}$ that mix quantum, gravitational, and relativistic structure. In the Quantum Lattice Model (QLM), these square-root forms are not taken as primitive. Instead, all Planck quantities admit a unified algebraic representation within a phase–action framework in which energy is defined covariantly as phase flow per unit proper time:

$$E = \hbar \frac{d\theta}{d\tau}. \tag{1.1}$$

Here θ is dimensionless phase (measured in radians) and τ is proper time along a worldline. Each lattice tick advances the physical phase by exactly one radian and transports precisely one quantum of reduced action:

$$\Delta\theta = 1, \qquad \Delta\tau = t_P, \qquad \Delta S = \hbar. \tag{1.2}$$

Angular frequency is therefore defined as

$$\omega \equiv \frac{d\theta}{d\tau}, \tag{1.3}$$

and cycle frequency is derived as $\nu = \omega/(2\pi)$. No full-cycle quantity is primitive in QLM; all foundational relations are defined at the per-radian level.

The fundamental *per-radian* Planck energy arises as the saturated single-tick realization of the phase–flow law. For one lattice tick,

$$E_P = \frac{\hbar}{t_P}. \tag{1.4}$$

The Planck energy therefore represents the per-tick phase throughput associated with the primitive action quantum. The full-cycle constant $h = 2\pi\hbar$ plays no foundational role in what follows.

Primitive spacings and exact SI structure. The QLM treats the Planck time t_P and Planck length ℓ_P as the primitive spacetime increments of a discrete four-dimensional lattice. These increments are linked by the exact SI identity

$$c = \frac{\ell_P}{t_P}. \tag{1.5}$$

This identity expresses the invariant null transport bound of the lattice and encodes the Lorentz-invariant relation between spatial and temporal increments in the Planck construction.

All numerical values in this paper use CODATA 2022 and the post-2019 SI, in which h, e, k_B, and c are defined constants. The electromagnetic constants ϵ_0, μ_0, and Z_0 inherit their uncertainties solely through the fine-structure constant α.

Unified algebraic structure of Planck, electromagnetic, and hydrogenic sectors. Once the covariant phase–flow law and the lattice transport identity $c = \ell_P/t_P$ are taken as primitive, Planck-scale action, vacuum electrodynamics, and the structure of the hydrogen atom admit a common reduced-action representation.

The Bohr identity

$$\hbar = m_e v_B a_0 \tag{1.6}$$

connects the reduced action quantum directly to the mechanical angular momentum of the hydrogen ground state. Combined with $c = \ell_P/t_P$, it gives the purely kinematic form of the fine-structure constant,

$$\alpha = \frac{v_B}{c}. \tag{1.7}$$

Within this framework, the Planck charge admits the equivalent set of identities

$$q_P^2 = \frac{4\pi\hbar}{Z_0} = \frac{2\pi\hbar}{Z_P^{(\mathrm{loop})}} = \frac{\hbar}{Z_P} = \hbar Y_P = 4\pi\epsilon_0\,\hbar c, \tag{1.8}$$

where the per-radian Planck impedance is

$$Z_P = \frac{Z_0}{4\pi}, \tag{1.9}$$

and the corresponding full-cycle (loop-aggregated) impedance is

$$Z_P^{(\mathrm{loop})} = 2\pi Z_P = \frac{Z_0}{2}, \qquad Z_P = \frac{Z_P^{(\mathrm{loop})}}{2\pi}. \tag{1.10}$$

These identities assemble into the electromagnetic quartet

$$\{V_P,\ I_P,\ P_P,\ Z_P\}, \tag{1.11}$$

which satisfies the algebraic closures

$$P_P = V_P I_P = \frac{V_P^2}{Z_P} = I_P^2 Z_P. \tag{1.12}$$

Loop-aggregated power relations involve $Z_P^{(\mathrm{loop})} = 2\pi Z_P$ when full 2π phase cycles are considered.

Scope of this work. This paper develops the complete Planck-unit foundation of the Quantum Lattice Model. Each canonical Planck quantity is first expressed in its traditional square-root form and then shown to collapse to its minimal QLM representation in terms of the primitive triplet ($\hbar$, ℓ_P, t_P). All results include CODATA-verified numerical evaluations and emphasize algebraic consistency across mechanical, electromagnetic, geometric, and hydrogenic sectors.

2 Constants and Conventions (CODATA 2022)

Table 1: Exact SI-defined constants and CODATA 2022 recommended values used throughout this paper. Derived Planck quantities are computed from these constants. Numerical values follow [23].

Quantity	Symbol	Value (CODATA 2022)
Speed of light (exact)	c	$2.997\,925 \times 10^{8}\,\mathrm{m\,s^{-1}}$
Planck constant (exact)	h	$6.626\,070 \times 10^{-34}\,\mathrm{J\,s}$
Reduced Planck constant (exact)	$\hbar$	$1.054\,572 \times 10^{-34}\,\mathrm{J\,s}$
Elementary charge (exact)	e	$1.602\,177 \times 10^{-19}\,\mathrm{C}$
Boltzmann constant (exact)	k_B	$1.380\,649 \times 10^{-23}\,\mathrm{J\,K^{-1}}$
Fine-structure constant	α	$7.297\,353 \times 10^{-3}$
Planck time (derived)	t_P	$5.391\,247 \times 10^{-44}\,\mathrm{s}$
Planck length (derived)	ℓ_P	$1.616\,255 \times 10^{-35}\,\mathrm{m}$
Electron mass	m_e	$9.109\,384 \times 10^{-31}\,\mathrm{kg}$
Bohr radius	a_0	$5.291\,772 \times 10^{-11}\,\mathrm{m}$
Vacuum impedance	Z_0	$3.767\,303 \times 10^{2}\,\Omega$
Vacuum permeability	μ_0	$1.256\,637 \times 10^{-6}\,\mathrm{N\,A^{-2}}$
Vacuum permittivity	ϵ_0	$8.854\,188 \times 10^{-12}\,\mathrm{F\,m^{-1}}$
Newtonian gravitational constant	G	$6.674\,300 \times 10^{-11}\,\mathrm{m^3\,kg^{-1}\,s^{-2}}$

3 Action Constants

The Quantum Lattice Model begins from the reduced–action phase relation

$$dS = \hbar\, d\theta, \tag{3.1}$$

which states that reduced action is transported in direct proportion to phase advance (measured in radians). Action is therefore quantized per radian rather than per full 2π cycle.

Dividing by proper time τ along a worldline defines energy as action flow per unit proper time:

$$E \equiv \frac{dS}{d\tau} = \hbar \frac{d\theta}{d\tau}, \tag{3.2}$$

which is the covariant phase–flow law of the QLM.

For a single saturated lattice tick,

$$\Delta\theta = 1, \qquad \Delta\tau = t_P, \tag{3.3}$$

so that

$$\Delta S = \hbar\, \Delta\theta = \hbar, \tag{3.4}$$

and the Planck energy follows as the maximal per–tick phase throughput,

$$E_P = \frac{\Delta S}{\Delta\tau} = \frac{\hbar}{t_P}. \tag{3.5}$$

Together with the Bohr angular–momentum relation

$$\hbar = m_e v_B a_0, \tag{3.6}$$

this establishes the complete per–radian action structure of the QLM. The full-loop action $h = 2\pi\hbar$ is derived and non-primitive, included only when converting to SI forms.

QLM Lattice Forms for $\hbar$ (Per–Radian Action)

Using the primitive per–radian identities

$$E_P = \frac{\hbar}{t_P}, \qquad \omega_P = \frac{1}{t_P}, \tag{3.7}$$

where ω_P denotes the Planck angular frequency ($\mathrm{rad\,s^{-1}}$), we obtain:

$$\hbar = E_P t_P, \tag{3.8}$$

$$\hbar = \frac{E_P}{\omega_P}. \tag{3.9}$$

The corresponding cycle frequency (Hz) is

$$\nu_P = \frac{\omega_P}{2\pi} = \frac{1}{2\pi t_P}, \tag{3.10}$$

which is derived and non-primitive.

An equivalent electromagnetic form follows from

$$q_P^2 = 4\pi\epsilon_0\hbar c, \qquad Z_0 = \frac{1}{\epsilon_0 c}, \tag{3.11}$$

giving

$$\hbar = \frac{Z_0}{4\pi}\, q_P^2 = Z_P\, q_P^2 = \frac{Z_P^{(\mathrm{loop})}}{2\pi}\, q_P^2, \tag{3.12}$$

where the per–radian Planck impedance is

$$Z_P = \frac{Z_0}{4\pi}, \tag{3.13}$$

and the loop–aggregated impedance is

$$Z_P^{(\mathrm{loop})} = 2\pi Z_P = \frac{Z_0}{2}. \tag{3.14}$$

Bohr Form (Explicit)

Hydrogen provides the direct angular–momentum interpretation:

$$\boxed{\hbar = m_e v_B a_0} \tag{3.15}$$

This identity is fully consistent with all QLM action forms and follows from the Bohr model of hydrogen [3, 4].

Numeric Validations (CODATA 2022)

Per–radian identity:

$$\begin{aligned}\hbar = E_P t_P &= \Big(1.956\,081 \times 10^{9}\,\mathrm{J}\Big)\Big(5.391\,247 \times 10^{-44}\,\mathrm{s}\Big)\\ &= 1.054\,572 \times 10^{-34}\,\mathrm{J\,s},\end{aligned} \tag{3.16}$$

consistent with CODATA 2022 [23].

Bohr form:

$$\begin{aligned}\hbar &= m_e v_B a_0\\ &= (9.109\,384 \times 10^{-31}\,\mathrm{kg})(2.187\,691 \times 10^{6}\,\mathrm{m\,s^{-1}})(5.291\,772 \times 10^{-11}\,\mathrm{m})\\ &= 1.054\,572 \times 10^{-34}\,\mathrm{J\,s},\end{aligned} \tag{3.17}$$

consistent with hydrogenic structure [3, 20].

Impedance–charge form (QLM).

$$\begin{aligned}\hbar &= \frac{Z_P^{(\mathrm{loop})}}{2\pi}\, q_P^2\\ &= \frac{1.883\,652 \times 10^{2}\,\Omega}{2\pi}\Big(3.517\,673 \times 10^{-36}\,\mathrm{C}^2\Big)\\ &= 1.054\,572 \times 10^{-34}\,\mathrm{J\,s},\end{aligned} \tag{3.18}$$

consistent with classical electrodynamics [18, 19].

Angular-frequency pairing:

$$\begin{aligned}\hbar &= \frac{E_P}{\omega_P}\\ &= \frac{1.956\,081 \times 10^{9}\,\mathrm{J}}{1.854\,858 \times 10^{43}\,\mathrm{s^{-1}}}\\ &= 1.054\,572 \times 10^{-34}\,\mathrm{J\,s},\end{aligned} \tag{3.19}$$

where $\omega_P = 1/t_P$.

Summary

$$\boxed{\hbar = E_P t_P = \frac{E_P}{\omega_P} = \frac{Z_0}{4\pi}\, q_P^2 = Z_P\, q_P^2 = \frac{Z_P^{(\mathrm{loop})}}{2\pi}\, q_P^2 = m_e v_B a_0} \tag{3.20}$$

$$\boxed{E = \frac{dS}{d\tau} = \hbar\,\frac{d\theta}{d\tau} \quad\Longrightarrow\quad E_P = \frac{\hbar}{t_P}} \tag{3.21}$$

$$\boxed{h = 2\pi\hbar} \tag{3.22}$$

These identities constitute the complete $\hbar$-based action structure of the Quantum Lattice Model. All subsequent Planck-unit derivations follow directly from the primitive covariant phase–flow law $E = \hbar\, d\theta/d\tau$, whose saturated single-tick realization yields $E_P = \hbar/t_P$.

4 Planck Lattice Foundations

4.1 Planck Time

The Planck time t_P is the fundamental temporal increment of the Quantum Lattice Model (QLM): one lattice tick corresponds to one radian of coherent phase evolution. In conventional dimensional analysis it is introduced through the reduced-action expression [1, 2]

$$t_P = \sqrt{\frac{\hbar G}{c^5}}, \tag{4.1}$$

but in the QLM this is not a definition. Instead, t_P is taken as the primitive temporal spacing of the lattice, and spatial and temporal increments are locked by the exact identity

$$c = \frac{\ell_P}{t_P}. \tag{4.2}$$

Spatial–Temporal Ratio of the Planck Lattice. A structural identity of the QLM is the fixed ratio between one spatial lattice link ℓ_P and one temporal tick t_P. A single phase hop per tick propagates at the invariant wave speed c, forcing the exact relation

$$\boxed{\frac{\ell_P}{t_P} = c \qquad \Longleftrightarrow \qquad \frac{t_P}{\ell_P} = \frac{1}{c}} \tag{4.3}$$

This identity encodes the invariant null transport relation of the discrete spacetime. Any slower advance corresponds to massive or bound configurations.

QLM form. From Eq. (4.2),

$$t_P = \frac{\ell_P}{c}, \tag{4.4}$$

showing that t_P is the temporal projection of the Planck phase-update cycle.

Collapse of the square-root form. To verify consistency, we show that the reduced-action expression (4.1) collapses exactly to the QLM identity (4.4). Starting from

$$t_P = \sqrt{\frac{\hbar G}{c^5}}, \tag{4.5}$$

we substitute for the product $\hbar G$ using the Planck-length relation

$$\ell_P^2 = \frac{\hbar G}{c^3} \qquad \Longrightarrow \qquad \hbar G = \ell_P^2 c^3. \tag{4.6}$$

Inserting Eq. (4.6) into Eq. (4.5) gives

$$t_P = \sqrt{\frac{\hbar G}{c^5}} \tag{4.7}$$

$$= \sqrt{\frac{\ell_P^2 c^3}{c^5}} \tag{4.8}$$

$$= \sqrt{\ell_P^2 \, \frac{c^3}{c^5}} \tag{4.9}$$

$$= \sqrt{\ell_P^2 \, \frac{1}{c^2}} \tag{4.10}$$

$$= \sqrt{\left(\frac{\ell_P}{c}\right)^2} \tag{4.11}$$

$$= \frac{\ell_P}{c}. \tag{4.12}$$

Thus the conventional dimensional form and the QLM lattice form are strictly identical:

$$t_P = \frac{\ell_P}{c}. \tag{4.13}$$

Bohr-anchored collapse. Using the hydrogenic identity [3]

$$\hbar = m_e v_B a_0, \tag{4.14}$$

and the QLM gravitational relation

$$G = \frac{\ell_P^5}{t_P^3 \, \hbar}, \tag{4.15}$$

Eq. (4.1) becomes

$$t_P = \sqrt{\frac{\hbar G}{c^5}} \tag{4.16}$$

$$= \sqrt{\frac{\ell_P^5}{t_P^3 c^5}} \tag{4.17}$$

$$= \sqrt{\frac{\ell_P^5}{t_P^3 (\ell_P^5 / t_P^5)}} \quad (\text{using } c = \ell_P / t_P)$$

$$= \sqrt{t_P^2} = t_P, \tag{4.18}$$

confirming algebraic equivalence of the hydrogenic, reduced-action, and QLM lattice forms.

Numeric evaluation (CODATA 2022).

$$t_P = \sqrt{\frac{(1.054\,572 \times 10^{-34}\,\mathrm{J\,s})(6.674\,300 \times 10^{-11}\,\mathrm{m^3\,kg^{-1}\,s^{-2}})}{(2.997\,925 \times 10^{8}\,\mathrm{m\,s^{-1}})^5}}$$
$$= 5.391\,247 \times 10^{-44}\,\mathrm{s}, \tag{4.19}$$

consistent with CODATA 2022 [23, 28].

$$t_P = \frac{1.616\,255 \times 10^{-35}\,\mathrm{m}}{2.997\,925 \times 10^{8}\,\mathrm{m\,s^{-1}}}$$
$$= 5.391\,247 \times 10^{-44}\,\mathrm{s}. \tag{4.20}$$

Thus the conventional square-root expression reduces to the QLM lattice identity

$$t_P = \frac{\ell_P}{c}, \tag{4.21}$$

demonstrating internal consistency of the QLM temporal construction.

4.2 Planck Length

The Planck length ℓ_P is the fundamental spatial increment of the Quantum Lattice Model. In conventional dimensional analysis it is introduced through the reduced-action expression [1, 2]

$$\ell_P = \sqrt{\frac{\hbar\, G}{c^3}}, \tag{4.22}$$

but in the QLM it is a *primitive* lattice spacing, linked directly to the Planck time by the exact lattice-velocity identity

$$c = \frac{\ell_P}{t_P}. \tag{4.23}$$

Thus the QLM definition is simply

$$\ell_P = c\, t_P, \tag{4.24}$$

identifying the spatial increment as the spatial projection of a single phase-update tick.

Collapse of the square-root form. To confirm consistency, we show that the reduced-action form (4.22) collapses exactly to the QLM identity (4.24).

Using the lattice form of the gravitational constant

$$G = \frac{\ell_P^5}{t_P^3\, \hbar}, \tag{4.25}$$

we substitute into Eq. (4.22):

$$\ell_P = \sqrt{\frac{\hbar G}{c^3}}. \tag{4.26}$$

Inserting Eq. (4.25) gives

$$\ell_P = \sqrt{\frac{\hbar\left(\frac{\ell_P^5}{t_P^3\hbar}\right)}{c^3}} \tag{4.27}$$

$$= \sqrt{\frac{\ell_P^5}{t_P^3 c^3}} \tag{4.28}$$

Now substitute the lattice kinematic identity

$$c = \frac{\ell_P}{t_P}, \tag{4.29}$$

so that

$$c^3 = \frac{\ell_P^3}{t_P^3}. \tag{4.30}$$

Using Eq. (4.30) in Eq. (4.28):

$$\ell_P = \sqrt{\frac{\ell_P^5}{t_P^3\left(\frac{\ell_P^3}{t_P^3}\right)}} \tag{4.31}$$

$$= \sqrt{\frac{\ell_P^5}{\ell_P^3}} \tag{4.32}$$

$$= \sqrt{\ell_P^2} \tag{4.33}$$

$$= \ell_P. \tag{4.34}$$

Equivalently, using Eq. (4.29),

$$\ell_P = c\,t_P. \tag{4.35}$$

Thus the conventional square-root expression is fully consistent with the QLM lattice identity.

Numeric evaluation (CODATA 2022). Using the QLM identity $\ell_P = c\,t_P$,

$$\begin{aligned}\ell_P &= (2.997\,925 \times 10^8\,\mathrm{m\,s^{-1}})(5.391\,247 \times 10^{-44}\,\mathrm{s}) \\ &= 1.616\,255 \times 10^{-35}\,\mathrm{m},\end{aligned} \tag{4.36}$$

in agreement with the CODATA 2022 recommended value [23, 29].

4.3 Planck Energy

In the Quantum Lattice Model, the Planck energy is not introduced through dimensional square roots, but arises from the covariant phase–flow law

$$E = \hbar \frac{d\theta}{d\tau}. \tag{4.37}$$

For a single saturated lattice tick,

$$\Delta\theta = 1, \qquad \Delta\tau = t_P, \tag{4.38}$$

so the Planck energy is the maximal per–tick phase throughput:

$$E_P = \frac{\hbar}{t_P}. \tag{4.39}$$

Thus E_P is not independently defined, but represents the saturated single-radian phase flow permitted by the lattice. All other Planck-energy expressions (including traditional 2π variants) are derived rather than primitive.

Bohr–QLM form. Using the Bohr identity [3, 4]

$$\hbar = m_e v_B a_0, \tag{4.40}$$

gives

$$E_P = \frac{m_e v_B a_0}{t_P}, \tag{4.41}$$

showing that the hydrogenic angular momentum provides a direct mechanical realization of the QLM phase–action quantum.

Numeric evaluation (CODATA 2022).

$$E_P = \frac{1.054\,572 \times 10^{-34}\,\mathrm{J\,s}}{5.391\,247 \times 10^{-44}\,\mathrm{s}} = 1.956\,081 \times 10^{9}\,\mathrm{J}, \tag{4.42}$$

consistent with CODATA 2022 [23, 28].

4.3.1 Square-Root Collapse of the Planck Energy

The conventional reduced-action definition introduced by Planck [1, 2] is

$$E_P = \sqrt{\frac{\hbar c^5}{G}}. \tag{4.43}$$

To demonstrate consistency with the QLM primitive identity $E_P = \hbar/t_P$, we eliminate G and c in favor of ℓ_P and t_P.

Using the standard Planck-length relation

$$\ell_P^2 = \frac{\hbar G}{c^3}, \tag{4.44}$$

we solve for G:

$$G = \frac{\ell_P^2 c^3}{\hbar}. \tag{4.45}$$

Substituting into Eq. (4.43) gives

$$E_P = \sqrt{\frac{\hbar c^5}{\frac{\ell_P^2 c^3}{\hbar}}} \tag{4.46}$$

$$= \sqrt{\frac{\hbar^2 c^2}{\ell_P^2}} \tag{4.47}$$

$$= \frac{\hbar c}{\ell_P}. \tag{4.48}$$

Now applying the QLM lattice identity

$$c = \frac{\ell_P}{t_P}, \tag{4.49}$$

we obtain

$$E_P = \frac{\hbar}{t_P}. \tag{4.50}$$

Thus the conventional square-root definition collapses exactly to the QLM per–radian identity.

4.3.2 Energy–Mass Relation Collapse

The relativistic identity $E = mc^2$ [6] must reproduce the QLM Planck energy when applied to the Planck mass.

The conventional definition of the Planck mass is

$$m_P = \sqrt{\frac{\hbar c}{G}}, \tag{4.51}$$

which collapses in QLM form to

$$m_P = \frac{\hbar t_P}{\ell_P^2}. \tag{4.52}$$

Using the lattice transport identity

$$c = \frac{\ell_P}{t_P}, \tag{4.53}$$

we obtain

$$E_P = m_P c^2 = \frac{\hbar t_P}{\ell_P^2}\left(\frac{\ell_P}{t_P}\right)^2 \tag{4.54}$$

$$= \frac{\hbar t_P}{\ell_P^2}\frac{\ell_P^2}{t_P^2}$$

$$= \frac{\hbar}{t_P}. \tag{4.55}$$

Thus the relativistic identity is fully consistent with the QLM per–radian Planck energy.

4.4 Planck Angular Frequency

In the Quantum Lattice Model, each lattice tick advances the coherent phase by exactly *one radian*. Therefore the fundamental temporal rate is the **Planck angular frequency**

$$\boxed{\omega_P = \frac{1}{t_P}} \tag{4.56}$$

with units of radians per second.

This is the only primitive temporal frequency in the QLM. Cycle frequency (Hz) is derived and given by

$$\nu_P = \frac{\omega_P}{2\pi} = \frac{1}{2\pi t_P}. \tag{4.57}$$

Energy–Frequency Correspondence

From the primitive phase–flow law

$$E = \hbar \frac{d\theta}{d\tau}, \tag{4.58}$$

energy is related to angular frequency by

$$E = \hbar\omega. \tag{4.59}$$

For a saturated lattice tick,

$$E_P = \hbar\omega_P = \frac{\hbar}{t_P}. \tag{4.60}$$

Collapse of the Square-Root Route

The conventional reduced-action angular frequency corresponding to Planck energy is

$$\omega_P = \sqrt{\frac{c^5}{\hbar G}}, \tag{4.61}$$

which follows from $E_P = \hbar\omega_P$ and the traditional Planck-energy expression [1, 2].

Using the standard Planck-length relation

$$\ell_P^2 = \frac{\hbar G}{c^3}, \tag{4.62}$$

we solve for G:

$$G = \frac{\ell_P^2 c^3}{\hbar}. \tag{4.63}$$

Substituting into Eq. (4.61) gives

$$\omega_P = \sqrt{\frac{c^5}{\hbar\left(\frac{\ell_P^2 c^3}{\hbar}\right)}}$$

$$= \sqrt{\frac{c^2}{\ell_P^2}} = \frac{c}{\ell_P}. \tag{4.64}$$

Using the QLM lattice identity

$$c = \frac{\ell_P}{t_P}, \tag{4.65}$$

we obtain

$$\omega_P = \frac{1}{t_P}, \tag{4.66}$$

recovering the QLM primitive identity (4.56).

Numeric (CODATA 2022)

$$\omega_P = \frac{1}{5.391\,247 \times 10^{-44}\,\mathrm{s}} = 1.854\,858 \times 10^{43}\,\mathrm{s}^{-1}, \tag{4.67}$$

$$\nu_P = \frac{1}{2\pi t_P} = 2.953\,250 \times 10^{42}\,\mathrm{Hz}. \tag{4.68}$$

Summary.

$$\boxed{\omega_P = \frac{1}{t_P}, \qquad E_P = \hbar\omega_P = \frac{\hbar}{t_P}, \qquad \nu_P = \frac{1}{2\pi t_P}.} \tag{4.69}$$

The Planck angular frequency ω_P is the only primitive temporal rate in the Quantum Lattice Model. Cycle frequency (Hz) is derived from angular frequency by division by 2π and plays no foundational role in the lattice structure.

4.5 Planck Wavenumber

In the Quantum Lattice Model the primitive spatial phase rate is the Planck angular wavenumber

$$\boxed{k_P = \frac{1}{\ell_P}} \tag{4.70}$$

with units of radians per meter.

This is the exact spatial analogue of the primitive angular frequency

$$\omega_P = \frac{1}{t_P}. \tag{4.71}$$

No additional 2π factor is required: both ω and k are defined per radian.

Wave–frequency closure. Using the primitive QLM identities

$$\omega_P = \frac{1}{t_P}, \tag{4.72}$$

$$c = \frac{\ell_P}{t_P}, \tag{4.73}$$

their ratio becomes

$$\frac{\omega_P}{k_P} = \frac{\frac{1}{t_P}}{\frac{1}{\ell_P}} = \frac{\ell_P}{t_P} = c. \tag{4.74}$$

Thus the fundamental QLM wave relation is

$$\boxed{\frac{\omega_P}{k_P} = c.} \tag{4.75}$$

This is the exact relativistic dispersion relation for null propagation in angular variables.

Numeric (CODATA 2022).

$$k_P = \frac{1}{1.616\,255 \times 10^{-35}\,\mathrm{m}} = 6.187\,066 \times 10^{34}\,\mathrm{m}^{-1}, \tag{4.76}$$

$$\omega_P = \frac{1}{5.391\,247 \times 10^{-44}\,\mathrm{s}} = 1.854\,858 \times 10^{43}\,\mathrm{s}^{-1}, \tag{4.77}$$

$$\frac{\omega_P}{k_P} = \frac{1.854\,858 \times 10^{43}\,\mathrm{s}^{-1}}{6.187\,066 \times 10^{34}\,\mathrm{m}^{-1}} = 2.997\,925 \times 10^{8}\,\mathrm{m\,s}^{-1} = c, \tag{4.78}$$

consistent with CODATA 2022 [23, 29, 28].

Summary.

$$\boxed{k_P = \frac{1}{\ell_P}, \qquad \omega_P = \frac{1}{t_P}, \qquad \frac{\omega_P}{k_P} = c.} \tag{4.79}$$

The spatial and temporal increments of the Planck lattice are therefore exactly matched. The invariant lattice wave speed c emerges directly from the angular phase-transport relation without introducing any additional 2π normalization.

4.6 Planck Mass

QLM definition (per–radian). In the Quantum Lattice Model the Planck mass follows from the saturated Planck energy

$$E_P = \frac{\hbar}{t_P}, \tag{4.80}$$

together with the relativistic inertial relation $E = mc^2$ [6]. Thus

$$m_P = \frac{E_P}{c^2} = \frac{\hbar}{t_P c^2}. \tag{4.81}$$

Using the lattice identity $c = \ell_P/t_P$, this becomes

$$m_P = \frac{\hbar}{t_P}\frac{t_P^2}{\ell_P^2} = \frac{\hbar\, t_P}{\ell_P^2}, \tag{4.82}$$

which is the fundamental QLM lattice form.

Collapse of the conventional square-root expression. The traditional Planck-mass expression [1, 2] is

$$m_P = \sqrt{\frac{\hbar c}{G}}, \tag{4.83}$$

which must reduce to Eq. (4.82).

Using the standard Planck-length relation

$$\ell_P^2 = \frac{\hbar G}{c^3}, \tag{4.84}$$

we solve for G:

$$G = \frac{\ell_P^2 c^3}{\hbar}. \tag{4.85}$$

Substituting into Eq. (4.83) gives

$$\begin{aligned} m_P &= \sqrt{\frac{\hbar c}{\frac{\ell_P^2 c^3}{\hbar}}} \\ &= \sqrt{\frac{\hbar^2}{\ell_P^2 c^2}} = \frac{\hbar}{\ell_P c}. \end{aligned} \tag{4.86}$$

Applying the lattice identity $c = \ell_P/t_P$,

$$m_P = \frac{\hbar}{\ell_P(\ell_P/t_P)} = \frac{\hbar t_P}{\ell_P^2}, \tag{4.87}$$

exactly matching the QLM definition.

Thus the QLM-native Planck mass is

$$\boxed{m_P = \frac{\hbar\, t_P}{\ell_P^2}} \tag{4.88}$$

Consistency with $E = mc^2$. Using $m_P = \hbar t_P/\ell_P^2$ and $c = \ell_P/t_P$,

$$m_P c^2 = \left(\frac{\hbar t_P}{\ell_P^2}\right)\left(\frac{\ell_P}{t_P}\right)^2 = \frac{\hbar}{t_P} = E_P, \tag{4.89}$$

confirming full consistency with the saturated Planck energy.

Bohr-anchored form. Using the hydrogenic identity [3] $\hbar = m_e v_B a_0$,

$$m_P = \frac{(m_e v_B a_0)\, t_P}{\ell_P^2}, \tag{4.90}$$

demonstrating that the Planck mass arises from the same reduced-action quantum governing the Bohr ground state.

Numeric (CODATA 2022).

$$k_P = \frac{1}{1.616\,255 \times 10^{-35}\,\mathrm{m}} = 6.187\,066 \times 10^{34}\,\mathrm{m}^{-1}, \tag{4.91}$$

$$\omega_P = \frac{1}{5.391\,247 \times 10^{-44}\,\mathrm{s}} = 1.854\,858 \times 10^{43}\,\mathrm{s}^{-1}, \tag{4.92}$$

$$\frac{\omega_P}{k_P} = \frac{1.854\,858 \times 10^{43}\,\mathrm{s}^{-1}}{6.187\,066 \times 10^{34}\,\mathrm{m}^{-1}} = 2.997\,925 \times 10^{8}\,\mathrm{m\,s}^{-1} = c, \tag{4.93}$$

consistent with CODATA 2022 [23, 29, 28].

Summary.

$$\boxed{m_P = \frac{\hbar}{t_P c^2} = \frac{\hbar\, t_P}{\ell_P^2} = \frac{m_e v_B a_0\, t_P}{\ell_P^2}} \tag{4.94}$$

All forms are algebraically identical in the QLM and represent the single per–radian Planck mass.

4.6.1 Planck Mass as Confinement Closure

Mass in the QLM may be interpreted as closed momentum inventory normalized by the lattice propagation speed:

$$m \equiv \frac{p_{\mathrm{loop}}}{c}. \tag{4.95}$$

Confinement of a quantum excitation to spatial scale r implies a characteristic momentum scale of order

$$p_{\mathrm{loop}} \sim \frac{\hbar}{r}, \tag{4.96}$$

consistent with the reduced Compton relation [37].

Thus

$$m = \frac{p_{\mathrm{loop}}}{c} \sim \frac{\hbar}{rc}. \tag{4.97}$$

Solving for r gives

$$r = \frac{\hbar}{mc}, \tag{4.98}$$

which is precisely the reduced Compton wavelength.

For the Planck mass,

$$r(m_P) = \frac{\hbar}{m_P c} = \ell_P, \tag{4.99}$$

showing that the Planck mass corresponds exactly to confinement at one lattice spacing.

The Planck mass therefore represents the self-consistent closure of the lattice under maximal spatial confinement, where the reduced Compton wavelength equals the primitive lattice spacing.

4.7 Planck Momentum

In the Quantum Lattice Model, the Planck momentum follows from the saturated Planck energy

$$p_P = \frac{E_P}{c}, \qquad E_P = \frac{\hbar}{t_P}, \tag{4.100}$$

together with the invariant lattice velocity. Using $c = \ell_P/t_P$ (Eq. (4.24)) gives

$$p_P = \frac{\left(\frac{\hbar}{t_P}\right)}{\left(\frac{\ell_P}{t_P}\right)} = \frac{\hbar}{\ell_P}, \tag{4.101}$$

which is the fundamental QLM lattice form.

Collapse of the conventional square-root expression. The traditional dimensional expression [1, 2] is

$$p_P = \sqrt{\frac{\hbar c^3}{G}}, \tag{4.102}$$

which must reduce to $\hbar/\ell_P$.

Substituting

$$c = \frac{\ell_P}{t_P}, \tag{4.103}$$

$$G = \frac{\ell_P^5}{t_P^3\,\hbar}, \tag{4.104}$$

gives

$$p_P = \sqrt{\frac{\hbar\left(\frac{\ell_P^3}{t_P^3}\right)}{\frac{\ell_P^5}{t_P^3\,\hbar}}} \tag{4.105}$$

$$= \sqrt{\frac{\hbar^2}{\ell_P^2}} = \frac{\hbar}{\ell_P}, \tag{4.106}$$

exactly matching the QLM identity.

Wavenumber form (primitive angular lattice). The QLM treats spatial phase rate per radian as primitive,

$$k_P = \frac{1}{\ell_P}, \tag{4.107}$$

so that the Planck momentum follows directly from the quantum relation $p = \hbar k$ [20]:

$$p_P = \hbar\, k_P = \frac{\hbar}{\ell_P}. \tag{4.108}$$

This is the spatial dual of the energy relation $E_P = \hbar/t_P$ and completes the (E_P, p_P) pair generated by the primitive lattice increments (t_P, ℓ_P). For null propagation the relativistic invariant relation $E^2 - (pc)^2 = 0$ holds identically [6].

Numeric verification (CODATA 2022). All independent routes give the same per–radian Planck momentum:

$$p_P = \frac{E_P}{c} = \frac{1.956\,081 \times 10^9\,\mathrm{J}}{2.997\,925 \times 10^8\,\mathrm{m\,s^{-1}}} = 6.524\,785 \times 10^1\,\mathrm{kg\,m\,s^{-1}}, \tag{4.109}$$

$$p_P = \frac{\hbar}{\ell_P} = \frac{1.054\,572 \times 10^{-34}\,\mathrm{J\,s}}{1.616\,255 \times 10^{-35}\,\mathrm{m}} = 6.524\,785 \times 10^1\,\mathrm{kg\,m\,s^{-1}}, \tag{4.110}$$

$$p_P = \hbar\, k_P = \left(1.054\,572 \times 10^{-34}\,\mathrm{J\,s}\right)\left(6.186\,390 \times 10^{34}\,\mathrm{m^{-1}}\right) = 6.524\,785 \times 10^1\,\mathrm{kg\,m\,s^{-1}}. \tag{4.111}$$

consistent with CODATA 2022 [23, 29, 28].

Summary.

$$\boxed{p_P = \frac{E_P}{c} = \frac{\hbar}{\ell_P} = \hbar\, k_P = \frac{m_e v_B a_0}{\ell_P}} \tag{4.112}$$

All expressions are algebraically identical and represent the unique per–radian Planck momentum of the Quantum Lattice Model.

4.8 Planck Force

Planck force follows from the saturated Planck energy

$$E_P = \frac{\hbar}{t_P}, \tag{4.113}$$

as the spatial rate at which this per–radian energy quantum is transported across one lattice spacing ℓ_P:

$$F_P = \frac{E_P}{\ell_P} = \frac{\hbar}{t_P\, \ell_P}. \tag{4.114}$$

Using the lattice identity $\ell_P = c\, t_P$ (Eq. (4.24)), this becomes

$$F_P = \frac{\hbar}{t_P^2\, c}. \tag{4.115}$$

Thus the Planck force represents the maximal per–radian momentum flux permitted by a single lattice tick.

Collapse of the conventional dimensional expression. The conventional Planck force [1, 2] is

$$F_P = \frac{c^4}{G}. \tag{4.116}$$

Using the standard Planck-length relation

$$\ell_P^2 = \frac{\hbar G}{c^3}, \tag{4.117}$$

we solve for G:

$$G = \frac{\ell_P^2 c^3}{\hbar}. \tag{4.118}$$

Substituting into Eq. (4.116) gives

$$\begin{aligned} F_P &= \frac{c^4}{\dfrac{\ell_P^2 c^3}{\hbar}} \\ &= \frac{\hbar c}{\ell_P^2}. \end{aligned} \tag{4.119}$$

Applying $c = \ell_P/t_P$,

$$F_P = \frac{\hbar(\ell_P/t_P)}{\ell_P^2} = \frac{\hbar}{\ell_P t_P}, \tag{4.120}$$

which exactly reproduces the QLM expression (4.114).

Thus the dimensional form collapses identically to the $\hbar$-based lattice expression.

Numeric verification (CODATA 2022). Using $E_P = \hbar/t_P$,

$$\begin{aligned} F_P = \frac{E_P}{\ell_P} &= \frac{1.956\,081 \times 10^9\,\mathrm{J}}{1.616\,255 \times 10^{-35}\,\mathrm{m}} \\ &= 1.210\,256 \times 10^{44}\,\mathrm{N}, \end{aligned} \tag{4.121}$$

in agreement with the invariant dimensional form

$$F_P = \frac{c^4}{G}, \tag{4.122}$$

consistent with CODATA 2022 [23, 29, 35].

Summary.

$$\boxed{F_P = \frac{E_P}{\ell_P} = \frac{\hbar}{t_P\,\ell_P} = \frac{\hbar}{t_P^2 c} = \frac{c^4}{G} = \frac{m_e v_B a_0\, c}{\ell_P^2}} \tag{4.123}$$

All forms are algebraically identical representations of the single per–radian Planck force in the Quantum Lattice Model.

4.9 Planck Acceleration

Planck acceleration represents the maximal coherent rate of change of lattice velocity across one Planck tick. Using the primitive lattice identity

$$c = \frac{\ell_P}{t_P}, \tag{4.124}$$

the QLM definition is

$$a_P = \frac{c}{t_P} = \frac{\ell_P}{t_P^2}, \tag{4.125}$$

which is the highest coherent acceleration permitted by the discrete phase–action lattice.

Collapse of the dimensional form. The traditional dimensional expression [1, 2] is

$$a_P = \sqrt{\frac{c^7}{\hbar\, G}}, \tag{4.126}$$

Using the standard Planck-length relation

$$\ell_P^2 = \frac{\hbar G}{c^3}, \tag{4.127}$$

we solve for G:

$$G = \frac{\ell_P^2 c^3}{\hbar}. \tag{4.128}$$

Substituting into Eq. (4.126) gives

$$\begin{aligned} a_P &= \sqrt{\frac{c^7}{\hbar\left(\frac{\ell_P^2 c^3}{\hbar}\right)}} \\ &= \sqrt{\frac{c^4}{\ell_P^2}} = \frac{c^2}{\ell_P}. \end{aligned} \tag{4.129}$$

Using $c = \ell_P/t_P$,

$$a_P = \frac{\left(\frac{\ell_P}{t_P}\right)^2}{\ell_P} = \frac{\ell_P}{t_P^2} = \frac{c}{t_P}, \tag{4.130}$$

identical to the QLM definition (4.125).

Thus the conventional square-root form collapses exactly to the primitive lattice identity.

Bohr-anchored form. Using the hydrogenic relation [3, 27]

$$v_B = \alpha c, \tag{4.131}$$

the Planck acceleration admits the equivalent form

$$a_P = \frac{c}{t_P} = \frac{v_B}{\alpha\, t_P}, \tag{4.132}$$

linking the Planck-scale acceleration directly to atomic kinematics.

Numeric verification (CODATA 2022).

$$a_P = \frac{2.997\,925 \times 10^8\,\mathrm{m\,s^{-1}}}{5.391\,247 \times 10^{-44}\,\mathrm{s}} = 5.560\,726 \times 10^{51}\,\mathrm{m\,s^{-2}}, \tag{4.133}$$

consistent with CODATA 2022 [23, 28].

Bohr-anchored check.

$$v_B = \alpha\, c = (7.297\,353 \times 10^{-3})\left(2.997\,925 \times 10^8\,\mathrm{m\,s^{-1}}\right) = 2.187\,691 \times 10^6\,\mathrm{m\,s^{-1}}, \tag{4.134}$$

$$a_P = \frac{v_B}{\alpha\, t_P} = \frac{2.187\,691 \times 10^6\,\mathrm{m\,s^{-1}}}{(7.297\,353 \times 10^{-3})\,5.391\,247 \times 10^{-44}\,\mathrm{s}} = 5.560\,726 \times 10^{51}\,\mathrm{m\,s^{-2}}, \tag{4.135}$$

in exact agreement with Eq. (4.133).

4.10 Planck Power

In the Quantum Lattice Model, the Planck power is the rate at which the saturated per–radian Planck energy is transported across one lattice tick. Since each tick carries one radian of phase and $\hbar$ of action,

$$E_P = \frac{\hbar}{t_P}, \tag{4.136}$$

and therefore

$$P_P = \frac{E_P}{t_P} = \frac{\hbar}{t_P^2}. \tag{4.137}$$

This is the unique primitive power scale in the QLM, because the lattice advances exactly one radian per tick.

Collapse of the dimensional form. The traditional reduced-action expression [1, 2] is

$$P_P = \frac{c^5}{G}. \tag{4.138}$$

Using the standard Planck-length relation

$$\ell_P^2 = \frac{\hbar G}{c^3}, \tag{4.139}$$

we solve for G:

$$G = \frac{\ell_P^2 c^3}{\hbar}. \tag{4.140}$$

Substituting into Eq. (4.138) gives

$$P_P = \frac{c^5}{\left(\frac{\ell_P^2 c^3}{\hbar}\right)} = \hbar\,\frac{c^2}{\ell_P^2}. \tag{4.141}$$

Using the lattice identity $c = \ell_P/t_P$,

$$P_P = \hbar \frac{\left(\frac{\ell_P}{t_P}\right)^2}{\ell_P^2} = \frac{\hbar}{t_P^2}, \tag{4.142}$$

identical to the QLM definition (4.137). Thus the dimensional expression collapses exactly to the $\hbar$-based lattice result.

Relation to Planck force. Using the $\hbar$-based Planck force

$$F_P = \frac{\hbar}{t_P \ell_P}, \tag{4.143}$$

and $c = \ell_P/t_P$, we find

$$P_P = F_P\, c = \frac{\hbar}{t_P \ell_P}\left(\frac{\ell_P}{t_P}\right) = \frac{\hbar}{t_P^2}, \tag{4.144}$$

again reproducing Eq. (4.137).

Energy-density representation. Using the Planck energy density

$$u_P = \frac{\hbar}{\ell_P^3 t_P}, \tag{4.145}$$

gives

$$P_P = u_P\, \ell_P^2\, c = \frac{\hbar}{t_P^2}, \tag{4.146}$$

corresponding to transporting one radian of action across one lattice face per tick.

Bohr-anchored representation. With $\hbar = m_e v_B a_0$ [3],

$$P_P = \frac{m_e v_B a_0}{t_P^2}, \tag{4.147}$$

showing again that the Planck–hydrogen correspondence arises from the same reduced-action quantum.

Numeric verification (CODATA 2022).

$$\begin{aligned} P_P = \frac{\hbar}{t_P^2} &= \frac{1.054\,572 \times 10^{-34}\,\mathrm{J\,s}}{(5.391\,247 \times 10^{-44}\,\mathrm{s})^2} \\ &= 3.628\,255 \times 10^{52}\,\mathrm{W}, \end{aligned} \tag{4.148}$$

consistent with CODATA 2022 [23, 28] and in agreement with the dimensional form c^5/G.

4.11 Planck Temperature

Definition (per–radian QLM form). In the Quantum Lattice Model, the Planck temperature follows from the saturated per–radian Planck energy

$$E_P = \frac{\hbar}{t_P}, \tag{4.149}$$

together with the Boltzmann relation $E = k_B T$ [2, 24]. Thus

$$T_P = \frac{E_P}{k_B} = \frac{\hbar}{t_P\, k_B}. \tag{4.150}$$

This is the unique thermal scale compatible with one radian of phase advance per lattice tick.

Collapse of the dimensional square-root form. The conventional reduced-action expression is

$$T_P = \sqrt{\frac{\hbar c^5}{G k_B^2}}. \tag{4.151}$$

Using the standard Planck-length identity

$$\ell_P^2 = \frac{\hbar G}{c^3}, \tag{4.152}$$

we solve for G:

$$G = \frac{\ell_P^2 c^3}{\hbar}. \tag{4.153}$$

Substituting into Eq. (4.151) gives

$$\begin{aligned} T_P &= \sqrt{\frac{\hbar c^5}{\left(\frac{\ell_P^2 c^3}{\hbar}\right) k_B^2}} \\ &= \sqrt{\frac{\hbar^2 c^2}{\ell_P^2 k_B^2}} = \frac{\hbar c}{\ell_P k_B}. \end{aligned} \tag{4.154}$$

Using the lattice identity $c = \ell_P / t_P$,

$$T_P = \frac{\hbar}{t_P k_B}, \tag{4.155}$$

identical to the QLM definition (4.150).

Bohr-anchored form. With the hydrogenic identity [3]

$$\hbar = m_e v_B a_0, \tag{4.156}$$

the Planck temperature becomes

$$T_P = \frac{m_e v_B a_0}{t_P\, k_B}. \tag{4.157}$$

Phase advance per Kelvin. Since a thermal increment ΔT corresponds to energy $E = k_B\,\Delta T$, the associated phase advance is

$$\vartheta = \frac{E}{E_P} = \frac{k_B\,\Delta T}{\hbar/t_P} = \Delta T\left(\frac{t_P\,k_B}{\hbar}\right). \tag{4.158}$$

Thus a temperature change of 1 K produces phase advance

$$\vartheta_{1\mathrm{K}} = \frac{1}{T_P} = \frac{t_P\,k_B}{\hbar}, \tag{4.159}$$

making temperature directly proportional to accumulated phase.

Numeric evaluation (CODATA 2022).

$$E_P = \frac{\hbar}{t_P} = \frac{1.054\,572\times10^{-34}\,\mathrm{J\,s}}{5.391\,247\times10^{-44}\,\mathrm{s}} = 1.956\,081\times10^{9}\,\mathrm{J}, \tag{4.160}$$

$$T_P = \frac{E_P}{k_B} = \frac{1.956\,081\times10^{9}\,\mathrm{J}}{1.380\,649\times10^{-23}\,\mathrm{J\,K^{-1}}} = 1.416\,784\times10^{32}\,\mathrm{K}, \tag{4.161}$$

$$\vartheta_{1\mathrm{K}} = \frac{1}{T_P} = 7.058\,239\times10^{-33}\,\mathrm{rad\,K^{-1}}, \tag{4.162}$$

consistent with CODATA 2022 [23, 28].

Summary.

$$\boxed{T_P = \frac{\hbar}{t_P\,k_B} = \frac{m_e v_B a_0}{t_P\,k_B}} \tag{4.163}$$

$$\boxed{\vartheta_{1\mathrm{K}} = \frac{1}{T_P} = \frac{t_P k_B}{\hbar}} \tag{4.164}$$

5 Geometric Foundations and Planck Densities of the QLM

5.1 Primitive Geometric Cells of the Lattice

In the Quantum Lattice Model the Planck length ℓ_P and Planck time t_P are the primitive spacetime increments of the discrete lattice. They define the fundamental geometric cells:

$$A_P = \ell_P^2, \qquad \mathcal{V}_P = \ell_P^3, \qquad \mathcal{V}_4 = \ell_P^3 t_P. \tag{5.1}$$

- A_P is the *Planck area*, the face of a lattice cell.
- $\mathcal{V}_P$ is the *Planck spatial volume*.
- $\mathcal{V}_4$ is the *Planck four–volume tick*, the minimal spacetime element updated each lattice tick.

Temporal–spatial locking. The lattice enforces the exact causal relation

$$c = \frac{\ell_P}{t_P}, \tag{5.2}$$

fixing the primitive angular phase rate

$$\omega_P = \frac{1}{t_P}, \qquad k_P = \frac{1}{\ell_P}, \tag{5.3}$$

where ω_P denotes angular frequency (radians per unit time).

The derived cycle frequency is

$$\nu_P = \frac{1}{2\pi t_P}, \tag{5.4}$$

and the loop wavenumber is

$$k_P^{(\mathrm{loop})} = \frac{2\pi}{\ell_P}. \tag{5.5}$$

The propagation rule follows identically in radian-normalized form:

$$\frac{\omega_P}{k_P} = c. \tag{5.6}$$

One radian per four–volume cell. Energy is defined covariantly as phase flow per unit proper time:

$$E = \hbar \frac{d\theta}{d\tau}. \tag{5.7}$$

For one lattice tick,

$$\Delta\theta = 1, \qquad \Delta\tau = t_P, \tag{5.8}$$

giving the saturated Planck energy

$$E_P = \frac{\hbar}{t_P}. \tag{5.9}$$

Each four–volume cell

$$\mathcal{V}_4 = \ell_P^3 t_P \tag{5.10}$$

therefore supports exactly one radian of coherent phase advance and one quantum of reduced action.

$$\boxed{\text{one radian of phase, one quantum of action } (\hbar) \text{ per } \mathcal{V}_4.} \tag{5.11}$$

5.2 Planck Phase Density (Primitive QLM Quantity)

Because each $\mathcal{V}_4 = \ell_P^3 t_P$ corresponds to a single radian of phase evolution, the intrinsic phase density is

$$\Phi_P = \frac{1}{\mathcal{V}_4} = \frac{1}{\ell_P^3 t_P}. \tag{5.12}$$

Using the primitive angular phase rate

$$\omega_P = \frac{1}{t_P}, \tag{5.13}$$

together with $\ell_P = c\,t_P$, we obtain the equivalent forms

$$\begin{aligned}\Phi_P &= \frac{\omega_P}{\ell_P^3} \\ &= \frac{1}{\ell_P^3 t_P} \\ &= \frac{1}{c^3 t_P^4}.\end{aligned} \tag{5.14}$$

If expressed in terms of the derived cycle frequency

$$\nu_P = \frac{1}{2\pi t_P}, \tag{5.15}$$

then

$$\Phi_P = \frac{2\pi\nu_P}{\ell_P^3}, \tag{5.16}$$

which is strictly a derived representation.

Numeric (CODATA 2022).

$$\Phi_P = \frac{1}{(1.616\,255 \times 10^{-35}\,\mathrm{m})^3 (5.391\,247 \times 10^{-44}\,\mathrm{s})} = 4.393\,202 \times 10^{147}\,\mathrm{m}^{-3}\,\mathrm{s}^{-1}, \tag{5.17}$$

consistent with CODATA 2022 [23, 29, 28].

5.3 Planck Energy Density (Action per Four–Volume)

Energy in the QLM is defined covariantly as phase flow per unit proper time:

$$E = \hbar \frac{d\theta}{d\tau}. \tag{5.18}$$

For one lattice tick,

$$\Delta\theta = 1, \qquad \Delta\tau = t_P, \tag{5.19}$$

so the saturated Planck energy is

$$E_P = \frac{\hbar}{t_P}. \tag{5.20}$$

The energy density associated with one Planck spatial cell is therefore

$$u_P = \frac{E_P}{\ell_P^3} = \frac{\hbar}{\ell_P^3 t_P} = \hbar\,\Phi_P. \tag{5.21}$$

Thus the Planck energy density is the reduced-action quantum multiplied by the intrinsic phase density of spacetime.

Numeric (CODATA 2022).

$$u_P = \frac{1.956\,081 \times 10^9\ \mathrm{J}}{(1.616\,255 \times 10^{-35}\ \mathrm{m})^3} = 4.632\,947 \times 10^{113}\ \mathrm{J\,m^{-3}}, \tag{5.22}$$

consistent with CODATA 2022 [23, 29, 28].

5.4 Planck Mass Density

Mass in the QLM follows from the relativistic relation

$$E = mc^2, \tag{5.23}$$

together with the saturated Planck energy

$$E_P = \frac{\hbar}{t_P}. \tag{5.24}$$

Using $c = \ell_P/t_P$, the Planck mass becomes

$$m_P = \frac{E_P}{c^2} = \frac{\hbar}{t_P}\frac{t_P^2}{\ell_P^2} = \frac{\hbar\, t_P}{\ell_P^2}, \tag{5.25}$$

which is the QLM lattice form.

The mass density of one Planck cell is therefore

$$\rho_P = \frac{m_P}{\ell_P^3} = \frac{\hbar\, t_P}{\ell_P^5}. \tag{5.26}$$

Collapse from dimensional form. The traditional expression [1, 2] is

$$\rho_P = \frac{c^5}{\hbar G^2}, \tag{5.27}$$

which, using

$$G = \frac{\ell_P^5}{t_P^3\,\hbar}, \qquad c = \frac{\ell_P}{t_P}, \tag{5.28}$$

collapses exactly to Eq. (5.26).

Energy–mass closure. Since $E_P = m_P c^2$ [6] and $\ell_P = ct_P$,

$$u_P = \rho_P\, c^2, \tag{5.29}$$

the relativistic identity holds identically in QLM units.

Numeric (CODATA 2022).

$$\rho_P = \frac{2.176\,434 \times 10^{-8}\ \mathrm{kg}}{(1.616\,255 \times 10^{-35}\ \mathrm{m})^3} = 5.154\,849 \times 10^{96}\ \mathrm{kg\,m^{-3}}, \tag{5.30}$$

consistent with CODATA 2022 [23, 29, 28].

5.5 Summary of Planck Densities

$$\boxed{\Phi_P = \frac{1}{\ell_P^3 t_P}} \qquad \boxed{u_P = \hbar\,\Phi_P} \qquad \boxed{\rho_P = \frac{\hbar t_P}{\ell_P^5}} \tag{5.31}$$

These quantities form the fundamental density hierarchy of the Quantum Lattice Model:

- Phase density Φ_P, determined by one radian per four–volume cell.
- Energy density $u_P = \hbar\,\Phi_P$, arising from the covariant phase–flow relation $E = \hbar\, d\theta/d\tau$.
- Mass density ρ_P, obtained from u_P through the relativistic identity $E = mc^2$.

All three densities descend from the primitive phase–action structure encoded in the Planck four–volume $\mathcal{V}_4 = \ell_P^3 t_P$.

6 Gravitational Impedance Equivalence

Gravity in the QLM is expressed in terms of the reduced action quantum $\hbar$ transported per radian of phase advance. Since energy is defined covariantly as

$$E = \hbar \frac{d\theta}{d\tau}, \tag{6.1}$$

all Planck-scale gravitational structure reduces, at the level of unit construction, to the primitive triplet $(\hbar, \ell_P, t_P)$ together with the invariant transport identity

$$c = \frac{\ell_P}{t_P}. \tag{6.2}$$

The fundamental QLM representation of the Newtonian gravitational constant is

$$\boxed{G = \frac{\ell_P^5}{t_P^3\,\hbar}} \tag{6.3}$$

and any equivalent Planck-unit representation of G reduces to Eq. (6.3) when expressed in reduced-action lattice units.

Equivalent Representations of G

Beginning with the standard Planck-time definition

$$t_P^2 = \frac{\hbar G}{c^5}, \tag{6.4}$$

introduced dimensionally by Planck [1, 2], solving for G yields

$$G = \frac{c^5 t_P^2}{\hbar}. \tag{6.5}$$

Likewise, using the standard Planck-length definition

$$\ell_P^2 = \frac{\hbar G}{c^3}, \tag{6.6}$$

solving for G gives the equivalent spatial form

$$G = \frac{c^3 \ell_P^2}{\hbar}. \tag{6.7}$$

Bohr-anchored temporal form. Using the hydrogenic identity

$$\hbar = m_e v_B a_0, \tag{6.8}$$

from the Bohr model [3, 4], the same expression becomes

$$G = \frac{c^5 t_P^2}{m_e v_B a_0}. \tag{6.9}$$

Pure lattice (length–time) form. Substituting $c = \ell_P/t_P$ into Eq. (6.5) gives

$$\begin{aligned} G &= \frac{\left(\frac{\ell_P}{t_P}\right)^5 t_P^2}{\hbar} \\ &= \frac{\ell_P^5}{t_P^3\,\hbar}, \end{aligned} \tag{6.10}$$

recovering the fundamental lattice form.

Equivalence of all forms. Since $c = \ell_P/t_P$, we have

$$\frac{c^5 t_P^2}{\hbar} = \frac{\ell_P^5}{t_P^3\,\hbar}. \tag{6.11}$$

Using $\hbar = m_e v_B a_0$, the Bohr-anchored form collapses identically. All representations therefore reduce uniquely to Eq. (6.3).

Numerical Verification (CODATA 2022)

Temporal form.

$$\begin{aligned} G &= \frac{c^5 t_P^2}{\hbar} \\ &= \frac{(2.997\,925 \times 10^8)^5 (5.391\,247 \times 10^{-44})^2}{1.054\,572 \times 10^{-34}} \\ &= 6.674\,300 \times 10^{-11}\,\mathrm{m}^3\,\mathrm{kg}^{-1}\,\mathrm{s}^{-2}, \end{aligned} \tag{6.12}$$

consistent with CODATA 2022 [23, 35].

Bohr-anchored form.

$$\begin{aligned} G &= \frac{c^5 t_P^2}{m_e v_B a_0} \\ &= \frac{(2.997\,925 \times 10^8)^5 (5.391\,247 \times 10^{-44})^2}{(9.109\,384 \times 10^{-31})(2.187\,691 \times 10^6)(5.291\,772 \times 10^{-11})} \\ &= 6.674\,300 \times 10^{-11}\,\mathrm{m}^3\,\mathrm{kg}^{-1}\,\mathrm{s}^{-2}, \end{aligned} \tag{6.13}$$

Pure lattice form.

$$\begin{aligned} G &= \frac{\ell_P^5}{t_P^3\,\hbar} \\ &= \frac{(1.616\,255 \times 10^{-35})^5}{(5.391\,247 \times 10^{-44})^3 (1.054\,572 \times 10^{-34})} \\ &= 6.674\,300 \times 10^{-11}\,\mathrm{m}^3\,\mathrm{kg}^{-1}\,\mathrm{s}^{-2}, \end{aligned} \tag{6.14}$$

in agreement with CODATA 2022.

Equivalence of GR, Schwarzschild, and Planck–Lattice Impedance

The gravitational redshift around a static mass M in general relativity is [8, 9, 13]

$$Z_g(r) = \left(1 - \frac{2GM}{c^2 r}\right)^{-1/2}. \tag{6.15}$$

Step 1: Schwarzschild-radius form. Defining the Schwarzschild radius

$$r_s = \frac{2GM}{c^2}, \tag{6.16}$$

gives

$$Z_g(r) = \left(1 - \frac{r_s}{r}\right)^{-1/2}. \tag{6.17}$$

Step 2: Convert to QLM Planck units. Using the lattice representation of G,

$$G = \frac{\ell_P^5}{t_P^3\,\hbar}, \tag{6.18}$$

together with $c = \ell_P/t_P$, we obtain

$$\begin{aligned} \frac{2GM}{c^2 r} &= \frac{2M}{r}\left(\frac{\ell_P^5}{t_P^3 \hbar}\right)\left(\frac{t_P^2}{\ell_P^2}\right) \\ &= 2\,\frac{M \ell_P^3}{\hbar t_P r}. \end{aligned} \tag{6.19}$$

Thus

$$Z_g(r) = \left[1 - 2\,\frac{M \ell_P^3}{\hbar t_P r}\right]^{-1/2}. \tag{6.20}$$

Step 3: Planck-mass reduction. With the Planck mass

$$m_P = \frac{\hbar t_P}{\ell_P^2}, \qquad \frac{1}{m_P} = \frac{\ell_P^2}{\hbar t_P},$$

we compute

$$2\,\frac{M\ell_P^3}{\hbar t_P r} = 2\left(\frac{M}{m_P}\right)\left(\frac{\ell_P}{r}\right).$$

Therefore

$$Z_g(r) = \left[1 - 2\left(\frac{M}{m_P}\right)\left(\frac{\ell_P}{r}\right)\right]^{-1/2}. \tag{6.21}$$

The Schwarzschild radius likewise becomes

$$\boxed{r_s = 2\,\frac{M\ell_P^3}{\hbar t_P} = 2\left(\frac{M}{m_P}\right)\ell_P} \tag{6.22}$$

Final Equivalence.

$$\boxed{Z_g(r) = \left(1 - \frac{2GM}{c^2 r}\right)^{-1/2} = \left(1 - \frac{r_s}{r}\right)^{-1/2} = \left(1 - 2\,\frac{M\ell_P^3}{\hbar t_P r}\right)^{-1/2}.} \tag{6.23}$$

The QLM gravitational impedance is therefore algebraically equivalent to the Schwarzschild redshift factor, expressed entirely in $\hbar$-based Planck lattice units.

7 Charge and Electromagnetism in the Quantum Lattice Model

Electromagnetism in the Quantum Lattice Model (QLM) descends from the covariant phase–flow law

$$E = \hbar\frac{d\theta}{d\tau}. \tag{7.1}$$

For one lattice tick,

$$\Delta\theta = 1, \qquad \Delta\tau = t_P, \tag{7.2}$$

giving the saturated Planck energy

$$E_P = \frac{\hbar}{t_P}. \tag{7.3}$$

Because the lattice is defined by the increments (ℓ_P, t_P) and the action per radian $\hbar$, all electromagnetic quantities must be algebraically determined by these primitives together with the vacuum response encoded in ϵ_0 and Z_0.

Planck Charge and Impedance

In SI units the Planck charge is defined by

$$q_P^2 = 4\pi\,\epsilon_0\,\hbar\,c, \tag{7.4}$$

which, using $Z_0 = 1/(\epsilon_0 c)$, can be written

$$q_P^2 = \frac{4\pi\hbar}{Z_0}. \tag{7.5}$$

The per–radian Planck impedance is therefore

$$Z_P = \frac{Z_0}{4\pi}, \tag{7.6}$$

so that

$$\hbar = Z_P\,q_P^2. \tag{7.7}$$

The loop–aggregated impedance is

$$Z_P^{(\mathrm{loop})} = 2\pi Z_P = \frac{Z_0}{2}. \tag{7.8}$$

7.1 The Planck Electromagnetic Quartet

Electromagnetic transport in the Quantum Lattice Model follows directly from the per–radian saturation identity

$$E_P = \frac{\hbar}{t_P}. \tag{7.9}$$

Voltage and current. Voltage is energy per unit charge:

$$V_P = \frac{E_P}{q_P} = \frac{\hbar}{t_P q_P}, \tag{7.10}$$

and current is charge transported per lattice tick:

$$I_P = \frac{q_P}{t_P}. \tag{7.11}$$

Impedance (per–radian). The Planck impedance follows from the ratio

$$Z_P = \frac{V_P}{I_P} = \frac{E_P t_P}{q_P^2} = \frac{\hbar}{q_P^2}. \tag{7.12}$$

Using the QLM charge identity

$$q_P^2 = \frac{\hbar}{Z_P}, \tag{7.13}$$

this reduces identically to

$$Z_P = \frac{Z_0}{4\pi}. \tag{7.14}$$

For a full 2π phase loop,

$$Z_P^{(\mathrm{loop})} = 2\pi Z_P = \frac{Z_0}{2}, \tag{7.15}$$

with

$$Z_P = \frac{Z_P^{(\mathrm{loop})}}{2\pi}. \tag{7.16}$$

Planck power (per–radian). The maximal electromagnetic power transported per lattice tick is

$$P_P = \frac{E_P}{t_P} = \frac{\hbar}{t_P^2}, \tag{7.17}$$

which satisfies

$$P_P = V_P I_P. \tag{7.18}$$

Equivalently,

$$V_P = \sqrt{P_P Z_P}, \qquad I_P = \sqrt{\frac{P_P}{Z_P}}. \tag{7.19}$$

Quartet structure. Thus the set

$$\boxed{\{Z_P,\ V_P,\ I_P,\ P_P\}} \tag{7.20}$$

forms the complete Planck electromagnetic quartet, defined per radian of coherent lattice phase advance.

All quantities descend algebraically from the primitive phase–flow law $E = \hbar\, d\theta/d\tau$ together with the impedance normalization $Z_P = Z_0/(4\pi)$. Loop–aggregated relations are obtained by replacing Z_P with $Z_P^{(\mathrm{loop})} = 2\pi Z_P$ when full 2π phase cycles are considered.

7.2 Fine–Structure Constant and Hydrogenic Calibration

QLM Interpretation of the Fine–Structure Constant.

The fine–structure constant is not a primitive parameter in the QLM. It emerges from the hydrogenic action identity together with the invariant lattice velocity.

Hydrogenic (Kinematic) Form

The Bohr action identity

$$\hbar = m_e v_B a_0, \tag{7.21}$$

gives

$$v_B = \frac{\hbar}{m_e a_0}. \tag{7.22}$$

Since the lattice speed is

$$c = \frac{\ell_P}{t_P}, \tag{7.23}$$

the fine–structure constant becomes

$$\boxed{\alpha = \frac{v_B}{c}.} \tag{7.24}$$

Thus α is the ratio between confined rotational motion and invariant lattice propagation.

Electromagnetic (Impedance) Form

In SI units,

$$\alpha = \frac{e^2}{4\pi\varepsilon_0\hbar c}. \tag{7.25}$$

Using

$$Z_0 = \frac{1}{\varepsilon_0 c}, \qquad Z_P = \frac{Z_0}{4\pi} = \frac{1}{4\pi\varepsilon_0 c}, \tag{7.26}$$

we obtain

$$\frac{1}{Z_P} = 4\pi\varepsilon_0 c. \tag{7.27}$$

Therefore

$$4\pi\varepsilon_0\hbar c = \frac{\hbar}{Z_P}. \tag{7.28}$$

Substituting into Eq. (7.25) gives

$$\boxed{\alpha = \frac{e^2 Z_P}{\hbar}.} \tag{7.29}$$

Solving for the elementary charge:

$$\boxed{e^2 = \alpha\frac{\hbar}{Z_P} = \alpha q_P^2,} \tag{7.30}$$

so that

$$\boxed{e = q_P\sqrt{\alpha}.} \tag{7.31}$$

Unified Interpretation

The hydrogenic velocity ratio and the electromagnetic impedance ratio are therefore equivalent:

$$\boxed{\frac{v_B}{c} = \frac{e^2 Z_P}{\hbar}.} \tag{7.32}$$

The fine–structure constant is thus a dimensionless scaling factor relating rotational confinement and electromagnetic admittance within the same reduced–action lattice framework.

Numeric (CODATA 2022).

$$\alpha = \frac{2.187\,691 \times 10^6\,\mathrm{m\,s^{-1}}}{2.997\,925 \times 10^8\,\mathrm{m\,s^{-1}}} = 7.297\,353 \times 10^{-3}, \tag{7.33}$$

in agreement with CODATA 2022.

8 Rydberg Scale in Bohr and Planck–Lattice Form

The Rydberg constant sets the fundamental spectral scale of hydrogen. In the QLM it arises from the hydrogenic action identity together with the invariant lattice velocity.

8.1 Hydrogenic form

The Bohr model gives

$$v_B = \alpha c, \qquad \hbar = m_e v_B a_0. \tag{8.1}$$

The standard spectroscopic form is

$$\boxed{R_\infty = \frac{\alpha}{4\pi a_0}.} \tag{8.2}$$

Using $v_B = \alpha c$ and $c = \ell_P/t_P$, this may be written

$$R_\infty = \frac{v_B t_P}{4\pi a_0 \ell_P}, \tag{8.3}$$

which is algebraically identical to Eq. (8.11).

Thus the Planck-lattice factors (ℓ_P, t_P) cancel, showing that the Rydberg scale is controlled by hydrogenic geometry rather than by the absolute Planck scale.

8.2 Angular (per–radian) form

The angular wavenumber is

$$k_\infty = 2\pi R_\infty = \frac{\alpha}{2a_0} = \frac{v_B}{2a_0 c}. \tag{8.4}$$

Numeric (CODATA 2022). Using the hydrogenic identity

$$R_\infty = \frac{\alpha}{4\pi a_0}, \tag{8.5}$$

with CODATA 2022 values α and a_0, we obtain

$$\begin{aligned} R_\infty &= \frac{\alpha}{4\pi a_0} \\ &= \frac{7.297\,353 \times 10^{-3}}{4\pi\,(5.291\,772 \times 10^{-11}\,\mathrm{m})} \\ &= 1.097\,373 \times 10^{7}\,\mathrm{m}^{-1}. \end{aligned} \tag{8.6}$$

The corresponding angular (per–radian) wavenumber is

$$\begin{aligned} k_\infty &= 2\pi R_\infty \\ &= 2\pi \left(1.097\,373 \times 10^{7}\,\mathrm{m}^{-1}\right) \\ &= 6.895\,000 \times 10^{7}\,\mathrm{m}^{-1}. \end{aligned} \tag{8.7}$$

The small difference between Eqs. (8.6)–(8.7) and the tabulated CODATA value reflects rounding of the displayed inputs (α and a_0); using full CODATA precision reproduces

$$R_\infty = 1.097\,373 \times 10^7\,\mathrm{m}^{-1}, \tag{8.8}$$

and therefore

$$k_\infty = 2\pi R_\infty = 6.895\,000 \times 10^7\,\mathrm{m}^{-1}, \tag{8.9}$$

in agreement with CODATA 2022 [23, 27, 31, 26].

8.3 Bohr form of the Rydberg frequency

The Rydberg frequency is defined by

$$\nu_R = c\,R_\infty. \tag{8.10}$$

The canonical hydrogenic expression for R_∞ is

$$R_\infty = \frac{\alpha^2 m_e c}{2h}, \tag{8.11}$$

so that

$$\nu_R = \alpha^2\,\frac{m_e c^2}{2h}. \tag{8.12}$$

Using

$$\alpha = \frac{v_B}{c}, \qquad h = 2\pi\hbar, \tag{8.13}$$

gives

$$\boxed{\nu_R = \frac{m_e v_B^2}{4\pi\,\hbar}} \tag{8.14}$$

This expression makes no *explicit* reference to charge or ϵ_0; the electromagnetic structure is encoded implicitly through the Bohr velocity v_B.

Energy interpretation. Since the Bohr kinetic energy is

$$E_B = \frac{1}{2} m_e v_B^2, \tag{8.15}$$

Eq. (8.14) may be written

$$\nu_R = \frac{E_B}{2\pi\hbar} = \frac{E_B}{h}. \tag{8.16}$$

Thus the Rydberg frequency is the Bohr kinetic energy expressed in units of full-loop action h.

Numerical validation (CODATA 2022).

$$v_B = 2.187\,691 \times 10^6\,\mathrm{m\,s^{-1}}, \qquad m_e = 9.109\,384 \times 10^{-31}\,\mathrm{kg}, \qquad \hbar = 1.054\,572 \times 10^{-34}\,\mathrm{J\,s}, \tag{8.17}$$

$$\nu_R = \frac{(9.109\,384 \times 10^{-31}\,\mathrm{kg})(2.187\,691 \times 10^6\,\mathrm{m\,s^{-1}})^2}{4\pi(1.054\,572 \times 10^{-34}\,\mathrm{J\,s})} \tag{8.18}$$

$$= 3.289\,842 \times 10^{15}\,\mathrm{Hz}, \tag{8.19}$$

in agreement with CODATA 2022:

$$\nu_R = 3.289\,842 \times 10^{15}\,\mathrm{Hz}. \tag{8.20}$$

[23, 27, 30, 26]

9 Planck Charge from Action and Impedance

In the QLM, electric charge is not an independent primitive. The Planck charge is fixed algebraically by the reduced action quantum $\hbar$ together with the electromagnetic impedance of the vacuum.

In SI electromagnetism,

$$q_P^2 = 4\pi\epsilon_0\,\hbar c, \tag{9.1}$$

and using

$$\epsilon_0 = \frac{1}{Z_0 c}, \tag{9.2}$$

this becomes

$$q_P^2 = \frac{4\pi\hbar}{Z_0}. \tag{9.3}$$

With the per–radian normalization

$$Z_P = \frac{Z_0}{4\pi}, \tag{9.4}$$

we obtain the compact QLM form

$$\boxed{q_P^2 = \frac{\hbar}{Z_P}.} \tag{9.5}$$

Equivalently,

$$\boxed{q_P = \sqrt{\frac{\hbar}{Z_P}}.} \tag{9.6}$$

Thus the Planck charge is completely determined by the action–impedance pair $(\hbar, Z_P)$ and is not an additional independent parameter of the model.

9.1 Vacuum Impedance, Admittance, Permittivity, and Permeability

The electromagnetic constants of free space can be expressed in geometric lattice form once the vacuum impedance Z_0 and the lattice velocity $c = \ell_P/t_P$ are specified.

From

$$Z_0 = \mu_0 c = \frac{1}{\epsilon_0 c}, \qquad c = \frac{\ell_P}{t_P}, \tag{9.7}$$

the permittivity and permeability become

$$\mu_0 = \frac{Z_0}{c} = Z_0 \frac{t_P}{\ell_P}, \qquad \epsilon_0 = \frac{1}{Z_0 c} = \frac{t_P}{Z_0\, \ell_P}. \tag{9.8}$$

Consistency check. Substituting these expressions gives

$$\mu_0 \epsilon_0 c^2 = 1, \tag{9.9}$$

showing that the Maxwell wave identity is fully consistent with the lattice relation $c = \ell_P/t_P$.

Impedance and Admittance Form

Using the per–radian normalization

$$Z_P = \frac{Z_0}{4\pi}, \qquad Y_P = \frac{1}{Z_P}, \tag{9.10}$$

the Planck charge identity

$$q_P^2 = \frac{\hbar}{Z_P}, \tag{9.11}$$

relates the vacuum constants to the action–impedance pair $(\hbar, Z_P)$.

Equivalently,

$$Z_0 = 4\pi Z_P = \frac{4\pi\hbar}{q_P^2}, \tag{9.12}$$

$$\mu_0 = \frac{Z_0}{c}, \tag{9.13}$$

$$\epsilon_0 = \frac{1}{Z_0 c}. \tag{9.14}$$

Thus the electromagnetic triad

$$(\epsilon_0,\ \mu_0,\ Z_0) \tag{9.15}$$

is algebraically determined once the vacuum impedance Z_0, the lattice velocity $c = \ell_P/t_P$, and the reduced action quantum $\hbar$ are specified.

9.2 Planck Electromagnetic Field Strengths

The fundamental (per–radian) Planck energy of the QLM lattice is

$$E_P = \frac{\hbar}{t_P}, \tag{9.16}$$

so the associated Planck energy density is

$$u_P = \frac{E_P}{\ell_P^3} = \frac{\hbar}{\ell_P^3 t_P} = \hbar\,\Phi_P, \tag{9.17}$$

representing the maximal coherent electromagnetic energy storable in a single Planck four–volume per radian of phase advance.

Electric and magnetic field amplitudes. Using the conventional electromagnetic energy–density relations,

$$u = \frac{1}{2}\epsilon_0\,\mathcal{E}^2 = \frac{\mathcal{B}^2}{2\mu_0}, \tag{9.18}$$

the per–radian QLM Planck electromagnetic fields are

$$\mathcal{E}_P = \sqrt{\frac{2u_P}{\epsilon_0}} = \sqrt{\frac{2\hbar}{\epsilon_0\,\ell_P^3\,t_P}}, \tag{9.19}$$

$$\mathcal{B}_P = \sqrt{2\mu_0\,u_P} = \sqrt{\frac{2\mu_0\,\hbar}{\ell_P^3\,t_P}}. \tag{9.20}$$

Wave–speed and impedance relations. Their ratio satisfies the plane–wave identity

$$\frac{\mathcal{E}_P}{\mathcal{B}_P} = \frac{1}{\sqrt{\epsilon_0\mu_0}} = c = \frac{\ell_P}{t_P}, \tag{9.21}$$

which is a direct consequence of the lattice kinematic identity.

Impedance enters through the magnetic field intensity $\mathcal{H}_P = \mathcal{B}_P/\mu_0$, giving

$$\frac{\mathcal{E}_P}{\mathcal{H}_P} = \sqrt{\frac{\mu_0}{\epsilon_0}} = Z_0, \tag{9.22}$$

so the standard vacuum impedance relation is recovered exactly at the Planck scale.

Numeric values (CODATA 2022). Using

$$\hbar = 1.054571817 \times 10^{-34}\ \mathrm{J\,s}, \quad \ell_P = 1.616255 \times 10^{-35}\ \mathrm{m}, \quad t_P = 5.391247 \times 10^{-44}\ \mathrm{s}, \tag{9.23}$$

$$\epsilon_0 = 8.8541878128 \times 10^{-12}\ \mathrm{F/m}, \quad \mu_0 = 1.25663706212 \times 10^{-6}\ \mathrm{N/A^2}, \tag{9.24}$$

the energy density is

$$u_P = 4.63294706 \times 10^{113}\ \mathrm{J/m^3}. \tag{9.25}$$

Thus the fundamental Planck electromagnetic fields are

$$\mathcal{E}_P = 3.23248064 \times 10^{62}\ \mathrm{V/m}, \tag{9.26}$$

$$\mathcal{B}_P = 1.07832016 \times 10^{53}\ \mathrm{T}. \tag{9.27}$$

Both satisfy

$$\frac{\mathcal{E}_P}{\mathcal{B}_P} = c, \qquad \frac{\mathcal{E}_P}{\mathcal{H}_P} = Z_0,$$

to machine precision.

9.3 Summary of Electromagnetic Closure

$$\boxed{\alpha = \frac{v_B}{c}} \qquad \boxed{q_P^2 = \frac{\hbar}{Z_P} = \hbar Y_P} \qquad \boxed{Z_P = \frac{Z_0}{4\pi}} \tag{9.28}$$

$$\boxed{V_P = \frac{E_P}{q_P}, \quad I_P = \frac{q_P}{t_P}, \quad P_P = \frac{\hbar}{t_P^2}} \tag{9.29}$$

$$\boxed{\mathcal{E}_P = \sqrt{\frac{2u_P}{\epsilon_0}}, \qquad \mathcal{B}_P = \sqrt{2\mu_0 u_P}} \tag{9.30}$$

Electromagnetism in the QLM therefore closes algebraically once the vacuum impedance Z_0 is specified. All electromagnetic quantities follow deterministically from the reduced action quantum $\hbar$, the lattice increments (ℓ_P, t_P), and the impedance normalization

$$Z_P = \frac{Z_0}{4\pi},$$

with loop–aggregated relations obtained by replacing Z_P by $Z_P^{(\mathrm{loop})} = 2\pi Z_P = Z_0/2$ when full 2π phase cycles are considered.

10 Bohr Radius (Consistency and QLM–Equivalent Forms)

Canonical and QLM–consistent forms. Using the Bohr momentum $p_B \equiv m_e v_B$ and the standard hydrogenic relation $v_B = \alpha c$ [3, 20], the Bohr radius may be written equivalently as

$$a_0 = \frac{\hbar}{m_e c\,\alpha} = \frac{\hbar}{m_e v_B} = \frac{\hbar}{p_B}. \tag{10.1}$$

Using the QLM primitive covariant phase–flow definition of energy,

$$E = \hbar\frac{d\theta}{d\tau}, \qquad \Delta\theta = 1,\ \Delta\tau = t_P \ \Rightarrow\ E_P t_P = \hbar, \qquad E_P = \frac{\hbar}{t_P}, \tag{10.2}$$

we may rewrite $\hbar$ as $E_P t_P$, giving the QLM-equivalent form

$$a_0 = \frac{E_P t_P}{m_e c\,\alpha}. \tag{10.3}$$

Applying the lattice kinematic identity $c = \ell_P/t_P$,

$$a_0 = \frac{E_P t_P^2}{m_e\,\alpha\,\ell_P}. \tag{10.4}$$

Equations (10.1), (10.3), and (10.4) are algebraically identical and differ only by QLM substitutions ($E_P = \hbar/t_P,\ c = \ell_P/t_P$).

Bohr identity $\Rightarrow \alpha = v_B/c$ (complete cancellation). The Bohr action identity [3]

$$\hbar = m_e v_B a_0 \tag{10.5}$$

inserted into Eq. (10.1) gives

$$a_0 = \frac{m_e v_B a_0}{m_e c\,\alpha} = a_0\,\frac{v_B}{c\alpha} \Rightarrow 1 = \frac{v_B}{c\alpha} \Rightarrow \boxed{\alpha = \frac{v_B}{c}}. \tag{10.6}$$

The same cancellation occurs if we start from the QLM form $a_0 = E_P t_P/(m_e c\alpha)$ with $E_P t_P = \hbar$.

Thus, in the QLM, the Bohr radius in the form (10.1) contains no *explicit* dependence on e, ϵ_0, or μ_0; the electromagnetic content is encoded implicitly through the dimensionless ratio α [27, 18].

Units checks. From (10.1):

$$[\hbar] = \mathrm{kg\,m^2\,s^{-1}}, \qquad [m_e c\alpha] = \mathrm{kg\,m\,s^{-1}}, \quad \Rightarrow \quad [a_0] = \mathrm{m}. \tag{10.7}$$

From (10.4):

$$[E_P t_P^2] = (\mathrm{J})(\mathrm{s}^2) = \mathrm{kg\,m^2}, \qquad [m_e \alpha \ell_P] = \mathrm{kg\,m}, \tag{10.8}$$

again giving $[a_0] = \mathrm{m}$.

Numerical validation (CODATA 2022). Using the canonical hydrogenic form

$$a_0 = \frac{\hbar}{m_e c\,\alpha}, \tag{10.9}$$

with CODATA 2022 values,

$$\begin{aligned} a_0 &= \frac{1.054\,572 \times 10^{-34}\,\mathrm{J\,s}}{(9.109\,384 \times 10^{-31}\,\mathrm{kg})(2.997\,925 \times 10^{8}\,\mathrm{m\,s^{-1}})(7.297\,353 \times 10^{-3})} \\ &= 5.291\,772 \times 10^{-11}\,\mathrm{m}, \end{aligned} \tag{10.10}$$

in agreement with the CODATA 2022 recommended value [23, 31, 27, 30].

Substituting the QLM identities $E_P = \hbar/t_P$ and $c = \ell_P/t_P$ reproduces the same result identically, confirming algebraic equivalence of the canonical and lattice forms.

Velocity confirmation. Using $\alpha = v_B/c$,

$$\begin{aligned} v_B = \alpha c &= (7.297\,353 \times 10^{-3})(2.997\,925 \times 10^{8}\,\mathrm{m\,s^{-1}}) \\ &= 2.187\,691 \times 10^{6}\,\mathrm{m\,s^{-1}}, \end{aligned} \tag{10.11}$$

consistent with CODATA 2022 [23, 27].

11 Elementary Charge (Closed Loop)

Planck charge from action and impedance. Using the QLM substitutions

$$\epsilon_0 = \frac{t_P}{Z_0 \ell_P}, \qquad c = \frac{\ell_P}{t_P}, \tag{11.1}$$

the Planck charge becomes

$$\begin{aligned} q_P^2 &= 4\pi\,\epsilon_0\,\hbar\,c \\ &= 4\pi\,\hbar \frac{t_P}{Z_0 \ell_P} \frac{\ell_P}{t_P} = \boxed{\frac{4\pi\hbar}{Z_0}}. \end{aligned} \tag{11.2}$$

Introducing the per–radian Planck impedance

$$Z_P = \frac{Z_0}{4\pi}, \tag{11.3}$$

gives the compact QLM identity

$$\boxed{q_P^2 = \frac{\hbar}{Z_P} = \hbar Y_P}, \qquad Y_P \equiv \frac{1}{Z_P}. \tag{11.4}$$

For a full 2π phase loop,

$$Z_P^{(\mathrm{loop})} = 2\pi Z_P = \frac{Z_0}{2}. \tag{11.5}$$

Elementary charge from fine–structure scaling. Since the fine–structure constant is defined by

$$\alpha = \frac{e^2}{q_P^2}, \tag{11.6}$$

we immediately obtain

$$\boxed{e^2 = \alpha\, q_P^2 = \alpha\, \frac{\hbar}{Z_P} = \alpha\, \hbar Y_P}. \tag{11.7}$$

Using $q_P^2 = 4\pi\epsilon_0\hbar c$, this is equivalently

$$e^2 = 4\pi\,\epsilon_0\,\hbar\,c\,\alpha, \tag{11.8}$$

which is the standard rearranged form of the SI definition of the fine–structure constant.

Thus the elementary charge is

$$\boxed{e = q_P\sqrt{\alpha}}. \tag{11.9}$$

Impedance form of the fine–structure constant. Using $q_P^2 = \hbar/Z_P$, the fine–structure constant takes the canonical QLM impedance form

$$\boxed{\alpha = \frac{e^2 Z_P}{\hbar} = \frac{e^2}{\hbar Y_P}}. \tag{11.10}$$

In loop–aggregated form,

$$\alpha = \frac{e^2 Z_P^{(\mathrm{loop})}}{2\pi\,\hbar}. \tag{11.11}$$

Units check.

$$[\epsilon_0] = \mathrm{C}/(\mathrm{V\,m}), \qquad [\hbar] = \mathrm{V\,C\,s}, \qquad [c] = \mathrm{m/s}, \tag{11.12}$$

$$[\epsilon_0 \hbar c] = \mathrm{C}^2, \qquad [e^2] = \mathrm{C}^2. \tag{11.13}$$

Numerical validation (CODATA 2022). Using

$$Z_0 = 376.730313668\ \Omega,$$

QLM route:

$$\begin{aligned} e^2 &= \alpha\,\frac{\hbar}{Z_P} \\ &= (7.2973525643 \times 10^{-3})(1.054571817 \times 10^{-34})\left(\frac{4\pi}{376.730313668}\right)\ \mathrm{C}^2 && (11.14) \\ &= 2.5669699665 \times 10^{-38}\ \mathrm{C}^2. && (11.15) \end{aligned}$$

Conventional route:

$$\begin{aligned} e^2 &= 4\pi\,\epsilon_0\,\hbar\,c\,\alpha \\ &= 4\pi(8.8541878128 \times 10^{-12})(1.054571817 \times 10^{-34})(2.99792458 \times 10^{8})(7.2973525643 \times 10^{-3})\ \mathrm{C}^2 && (11.16) \\ &= 2.5669699665 \times 10^{-38}\ \mathrm{C}^2. && (11.17) \end{aligned}$$

Exact SI value:

$$e^2_{\mathrm{exact}} = (1.602176634 \times 10^{-19})^2 = 2.566969966535569956 \times 10^{-38}\ \mathrm{C}^2. \tag{11.18}$$

Both derived expressions agree with the exact SI value to the displayed precision. Any residual numerical difference arises from finite rounding of the displayed constants; the only physical uncertainty enters through the measured fine–structure constant α [23, 27].

12 Conclusion

The Quantum Lattice Model (QLM) reconstructs Planck-scale physics from a single deterministic covariant phase–action postulate:

$$E = \hbar \frac{d\theta}{d\tau}, \qquad c = \frac{\ell_P}{t_P}. \tag{12.1}$$

For a single coherent lattice tick $\Delta\theta = 1,\ \Delta\tau = t_P$, this primitive phase–flow law yields the saturation identity

$$E_P = \frac{\hbar}{t_P}, \qquad E_P t_P = \hbar. \tag{12.2}$$

With $(\hbar, \ell_P, t_P)$ taken as primitive lattice increments, all canonical Planck quantities follow algebraically. Mass, momentum, energy density, force, electromagnetic charge, impedance, field amplitudes, and power reduce to minimal expressions containing no additional dimensional structure.

Hydrogen provides the empirical calibration of this framework. The Bohr identities,

$$\hbar = m_e v_B a_0, \qquad \alpha = \frac{v_B}{c}, \tag{12.3}$$

demonstrate that the fine–structure constant is a geometric phase–velocity ratio linking rotational atomic motion to linear lattice propagation. Atomic observables therefore anchor the Planck lattice without introducing new dimensional parameters.

Electromagnetism closes consistently within the same phase–action structure. The Planck charge is fixed by the vacuum impedance normalization, and the electromagnetic quartet

$$(Z_P,\ V_P,\ I_P,\ P_P) \tag{12.4}$$

is uniquely determined by the per–radian saturation identity $E_P t_P = \hbar$. Vacuum permittivity, permeability, and impedance follow from the lattice spacetime increments together with the CODATA 2022 constants.

Across mechanical, electromagnetic, and hydrogenic sectors, all quantities collapse to a single internally consistent algebraic framework based on $(\hbar, \ell_P, t_P)$ and experimentally measured dimensionless ratios. No additional dimensional parameters beyond the CODATA constants enter at the level of Planck-unit construction.

The present work establishes the minimal canonical foundation of the Quantum Lattice Model. Subsequent developments apply this deterministic lattice structure to gravitational impedance, atomic structure, and cosmological dynamics.

Fundamental Constants and QLM Planck Units

Quantity	**Symbol**	**Definition**	**Value / Status**
Exact SI-Defined Constants (Post-2019)			
Speed of light	c	defined	2.99792458×10^{8} m/s (exact)
Planck constant	h	defined	$6.62607015 \times 10^{-34}$ J · s (exact)
Reduced Planck const.	$\hbar$	$h/(2\pi)$	exact (derived from h)
Elementary charge	e	defined	$1.602176634 \times 10^{-19}$ C (exact)
CODATA 2022 Measured Constants			
Gravitational constant	G	measured	6.67430×10^{-11} $\mathrm{m^3kg^{-1}s^{-2}}$
Fine-structure const.	α	measured	$7.2973525643 \times 10^{-3}$
Vacuum impedance	Z_0	$\mu_0 c$	$3.76730313668 \times 10^{2}$ Ω
Vacuum permittivity	ϵ_0	$1/(Z_0 c)$	$8.8541878128 \times 10^{-12}$ F/m
Vacuum permeability	μ_0	Z_0/c	$1.25663706212 \times 10^{-6}$ $\mathrm{N/A^2}$
Derived Planck / QLM Quantities (Computed)			
Planck time	t_P	$\sqrt{\hbar G/c^5}$	5.391247×10^{-44} s
Planck angular frequency	ω_P	$1/t_P$	$1.85485843999 \times 10^{43}$ $\mathrm{s^{-1}}$
Planck length	ℓ_P	ct_P	1.616255×10^{-35} m
Planck wavenumber (per–rad.)	k_P	$1/\ell_P$	$6.187065997 \times 10^{34}$ $\mathrm{m^{-1}}$
Planck energy	E_P	$\hbar/t_P$	$1.95608143653 \times 10^{9}$ J
Planck mass	m_P	$\hbar t_P/\ell_P^2$	2.176434×10^{-8} kg
Planck impedance (per–rad.)	Z_P	$Z_0/(4\pi)$	$2.9979245800 \times 10^{1}$ Ω
Planck admittance	Y_P	$1/Z_P$	$3.33564095198 \times 10^{-2}$ S
Planck charge	q_P	$\sqrt{\hbar/Z_P}$	$1.87554595641 \times 10^{-18}$ C

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